



Project Report

SameNeed - Connecting People with Shared Service Needs

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Introduction

SameNeed is a Java-based web application designed to help customers who require the same local service discover each other, form a service group, and collectively coordinate with a service provider.

The central idea of SameNeed is that customers do not need to have the exact same problem. They only need to require the **same service**.

For example, one customer may have a washing machine that is not draining, while another may have a leakage problem. Their individual problems are different, but both require **Washing Machine Repair**. SameNeed allows such customers to become part of the same service group when their locality and preferred schedule are compatible.

After a group is formed, members can communicate through a private real-time chat. A suitable service provider can view the group's requirements and submit a group-based price offer. The group coordinator can negotiate through counter-offers, after which individual members can confirm or decline participation. Once the required confirmations are received, a booking can be created.

The application also provides separate portals for customers, service providers, and administrators.

Problem Statement

Customers usually arrange local services such as appliance repair, plumbing, electrical work, or servicing individually. When several customers in the same locality require the same service around a similar period, they generally have no simple mechanism to discover one another, form a group, and collectively approach a service provider.

Existing service-booking systems generally process customer requests independently. They do not provide a dedicated workflow for grouping customers based on their common service requirement and supporting collective negotiation.



For example:

- Customer A needs washing machine repair because the machine is not draining.
- Customer B needs washing machine repair because of water leakage.
- Customer C needs washing machine repair because the machine is making unusual noise.

Their problems are different, but their required service is the same.

SameNeed addresses this problem by allowing compatible customers to discover one another, form a group, communicate anonymously, receive provider offers, negotiate collectively, and proceed toward a group booking.

The platform does not guarantee a discount. Instead, it creates an opportunity for customers and service providers to negotiate a group-based price.

organize their academic journey.

Functional Requirements

3.1 Customer Module

The system shall allow customers to:

1. Register an account.
2. Log in and log out.
3. Manage their profile.
4. Create service requests.
5. Select a service category and service.
6. Specify their individual problem.
7. Specify locality and preferred date/time.
8. Specify an optional expected budget.
9. Browse and search available requests.
10. Join compatible service groups.
11. Leave groups where permitted.
12. View anonymous group members.
13. Communicate through real-time group chat.
14. View provider offers.
15. View negotiation history.
16. Respond to available offers.
17. View and track bookings.



18. Receive in-app notifications.
19. Submit reviews after completed services.
20. Report issues or disputes.

3.2 Service Provider Module

The system shall allow service providers to:

1. Register and log in.
2. Manage their provider profile.
3. Add supported services.
4. Define service areas.
5. View eligible service groups.
6. View aggregated group requirements.
7. View individual problems reported by group members.
8. Submit group-based price offers.
9. Participate in counter-offer negotiation.
10. View booking information.
11. Update service status.
12. View service history.
13. Receive customer reviews.

3.3 Administrator Module

The system shall allow administrators to:

1. Log in securely.
2. View application statistics.
3. Manage customers.
4. Manage service providers.
5. Verify service providers.
6. Manage service categories.
7. Manage services.
8. Monitor service requests.
9. Monitor bookings.
10. Handle reports and disputes.
11. Manage user accounts.
12. Suspend accounts when required.
13. Export booking information as CSV.
14. Access the restricted entity metadata functionality.



3.4 Group and Negotiation Requirements

The system shall:

- Match customers based on the same service requirement.
- Consider locality and compatible scheduling information.
- Allow different individual problems within the same service group.
- Maintain request-specific anonymous member identities.
- Restrict group size to the configured maximum.
- Maintain a negotiation history.
- Support provider offers and counter-offers.
- Support individual member responses after coordinator acceptance.
- Handle offer expiry.

3.5 Communication Requirements

The system shall provide:

- Request-specific group chat.
- Real-time message delivery through WebSocket.
- Persistent storage of chat messages.
- Access control so users can only access groups they are authorized to participate in.

Non-Functional Requirements

4.1 Security

- Passwords shall be stored using BCrypt hashing.
- Protected resources shall require authentication.
- Role-based access shall restrict customer, provider, and administrator functionality.
- HTTP sessions shall be used for authentication.
- Personal information shall not be unnecessarily exposed to group members.
- Anonymous aliases shall be used during group formation.

4.2 Performance

- Database queries should retrieve only the required records.
- Group matching should filter requests using relevant service and locality information.



- WebSocket communication should avoid unnecessary page refreshes.
- Background processing should run periodically rather than continuously consuming resources.

4.3 Reliability

- Important application data shall be persisted in MySQL.
- Chat messages and negotiation records shall remain available after page refresh.
- Invalid operations shall return appropriate errors.
- Expired offers shall be handled automatically.

4.4 Usability

- The interface should be understandable to first-time users.
- Customer, provider, and administrator functions are separated.
- Forms should provide validation feedback.
- Important statuses such as booking and offer status should be clearly displayed.

4.5 Maintainability

- Application functionality is divided into controllers, services, DAOs, models, filters, utilities, and configuration components.
- Database access is separated from request-handling logic.
- Enumerations are used for fixed states such as roles and booking statuses.
- Configuration values are kept outside the main application logic where appropriate.

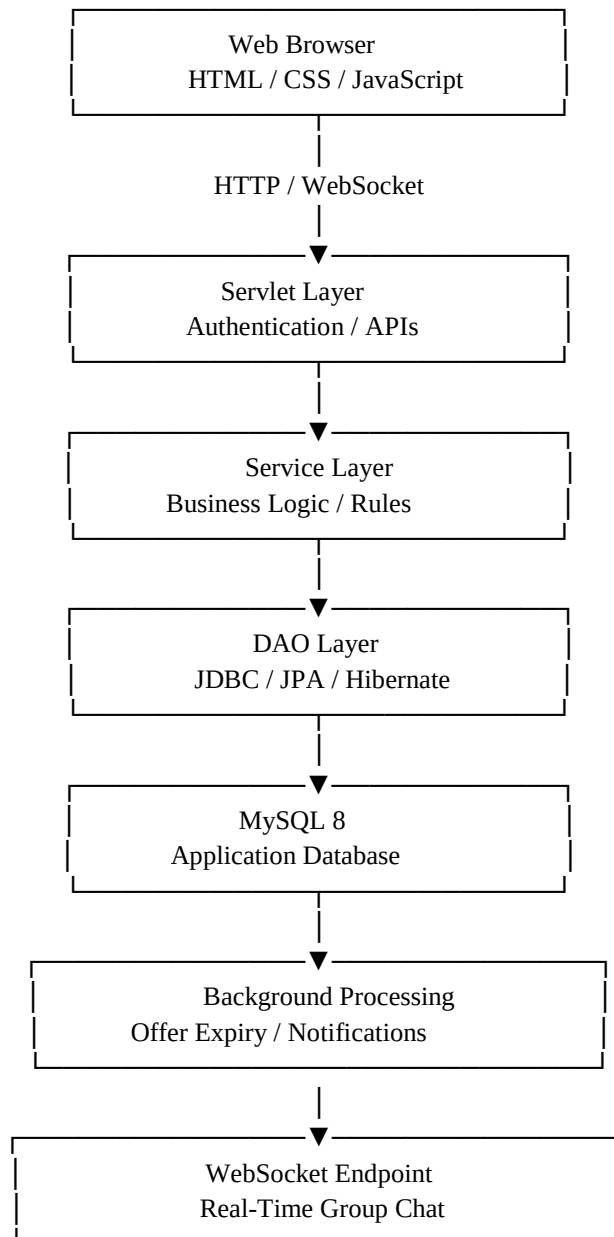
4.6 Scalability

The application uses modular components and database-backed persistence so that additional services, providers, customer groups, and application features can be added without redesigning the complete system.



System Architecture

SameNeed follows a layered Java web application architecture.





Architecture Layers

Presentation Layer

The frontend uses plain HTML, CSS, and JavaScript. JavaScript communicates with the backend through HTTP APIs and WebSocket connections.

Controller Layer

Jakarta Servlets receive HTTP requests, validate basic request information, invoke application services, and return JSON responses.

Service Layer

The service layer contains the main business rules such as request creation, group joining, offer negotiation, booking processing, and notification handling.

Data Access Layer

The DAO layer handles database operations using JDBC and selected JPA/Hibernate components.

Database Layer

MySQL stores users, service requests, groups, offers, negotiations, bookings, messages, reviews, notifications, reports, and related data.

Design Decisions & Rationale

6.1 Use Case Diagram

The major actors are:

- Customer
- Service Provider
- Administrator



Customer Use Cases

- Register/Login
- Create Service Request
- Browse Requests
- Join Group
- Chat with Group
- View Offers
- Respond to Offers
- View Booking
- Submit Review
- Report Issue

Service Provider Use Cases

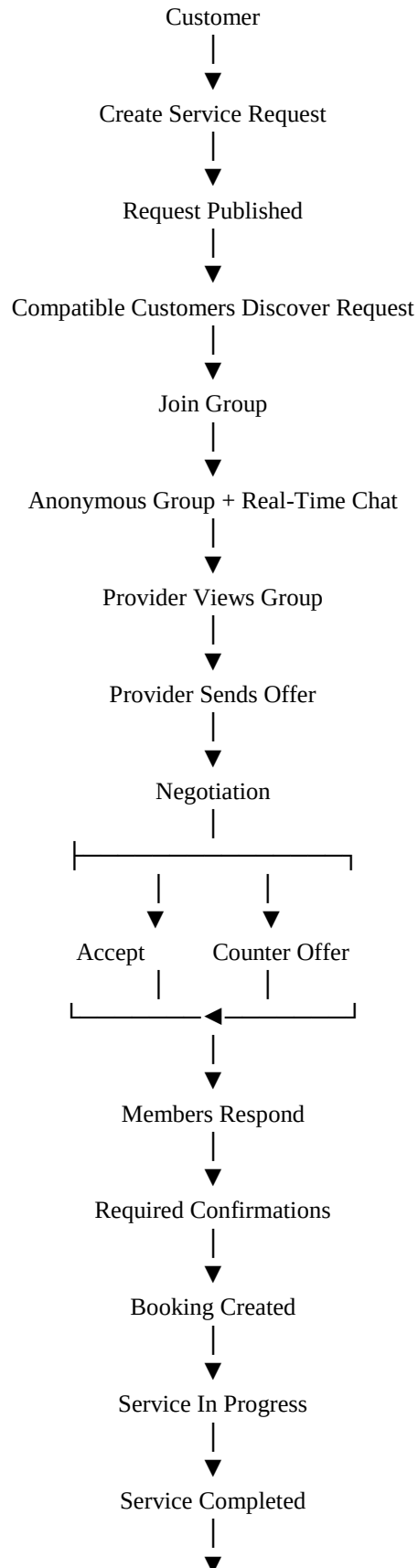
- Register/Login
- Manage Profile
- Manage Services
- View Service Groups
- View Requirements
- Submit Offer
- Negotiate
- Manage Booking
- Update Service Status
- View Reviews

Administrator Use Cases

- Login
- Manage Users
- Verify Providers
- Manage Services
- Monitor Requests
- Monitor Bookings
- Manage Reports
- Manage Disputes
- Export Data

6.2 Workflow Diagram

The main workflow is:

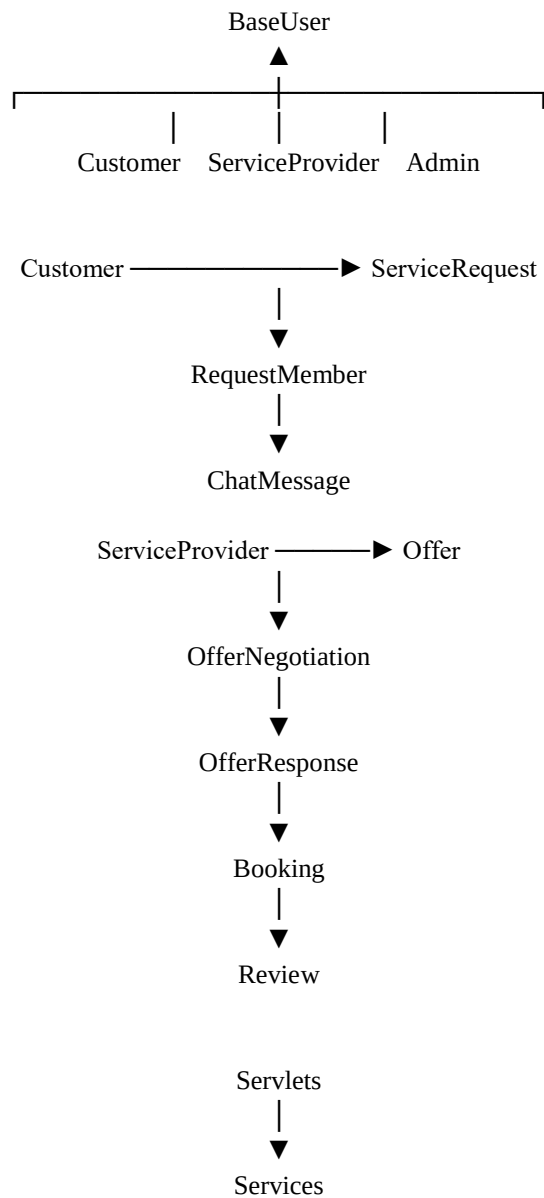


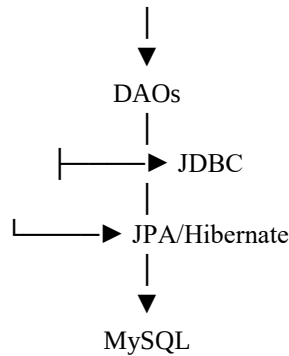


Review

6.3 Class / Component Diagram

The major components are:





Design Decisions & Rationale

7.1 Jakarta Servlets Instead of a Heavy Frontend Framework

Plain HTML, CSS, and JavaScript were selected because the project focuses on Java web development and server-side concepts. This also keeps the frontend lightweight and easier to understand.

7.2 MySQL for Persistent Storage

MySQL was selected because SameNeed contains relational data with relationships between users, services, requests, groups, offers, bookings, and reviews.

7.3 JDBC and JPA/Hibernate

JDBC is used for most application persistence to provide direct database interaction and demonstrate JDBC concepts.

JPA/Hibernate is used for selected entities to demonstrate ORM concepts and provide practical experience with entity mapping and persistence.

Using both approaches also allows comparison between direct SQL-based persistence and ORM-based persistence.

7.4 Anonymous Group Members

A request-specific anonymous alias protects customer identity during group formation.

Instead of showing a member's actual name, the system can display:

Group Creator
Member 02



Member 03

The actual user identity remains associated internally with the membership record.

7.5 Service-Based Matching

The system does not attempt to compare the exact wording of customer problems.

Matching is based on structured information such as:

- Service
- Locality
- Preferred date
- Compatible time
- Request status
- Group capacity

This keeps the matching logic deterministic and understandable.

7.6 WebSocket for Group Chat

HTTP requests would require repeated polling or page refreshes to obtain new messages. WebSocket provides a persistent communication channel for real-time group messaging.

7.7 Background Offer Expiry

A background worker periodically checks offers whose validity period has expired and updates their status. This demonstrates Java multithreading while solving an actual application requirement.

7.8 Role-Based Access

Different users have different responsibilities. Authentication and role-based filters prevent users from accessing functions that belong to another role.

7.9 Configurable Group Capacity

The group capacity is treated as a configurable application rule instead of being repeated as a fixed number throughout the code. The default maximum is 10 members including the creator.



Implementation Details

8.1 Backend

The backend is implemented using Java 17 and Jakarta Servlet 6.0.

Major backend components include:

- Authentication servlets
- Request management
- Group membership management
- Offer management
- Booking management
- Review management
- Notification management
- Admin management
- Report management
- CSV export
- Reflection-based metadata functionality

8.2 Authentication

Users authenticate through the login system.

Passwords are processed using BCrypt before being stored.

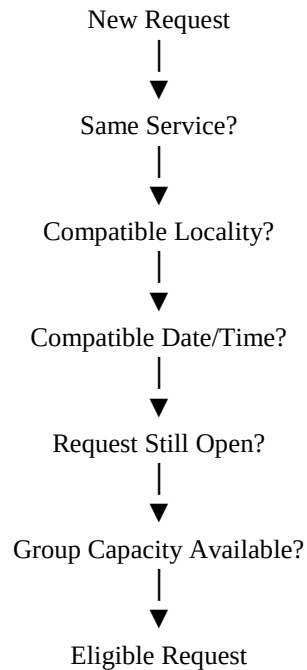
HTTP sessions are used to maintain authenticated user state.

8.3 Request Matching



The request-matching logic considers structured attributes rather than free-text similarity.

A simplified matching process is:



8.4 Group Management

Each service request can have a corresponding group.

A membership record connects a user to the request and contains the user's request-specific anonymous alias.

The system prevents duplicate membership and prevents groups from exceeding their configured capacity.

8.5 Offer and Negotiation

Offers are stored with information such as:

- Provider
- Request
- Amount per member
- Required members
- Validity period
- Status



- Creation time

Counter-offers are stored as negotiation records rather than replacing the previous offer.

This creates a persistent history of the negotiation.

8.6 WebSocket Chat

The group chat uses Jakarta WebSocket.

The WebSocket endpoint is scoped to a particular service request, allowing authorized members of that group to communicate in real time.

Messages are also stored in MySQL so that they can be retrieved after reconnecting or refreshing the page.

8.7 Booking

A booking is created after the offer and member participation conditions are satisfied.

The booking tracks the service provider, participating group/request, agreed price, and service status.

8.8 Notifications

Application events generate in-app notifications.

The notification dispatcher processes pending notification tasks and stores notification records for users to view through the application.

8.9 Offer Expiry

The offer expiry worker periodically checks pending offers.

When an offer passes its validity time, it is marked as expired and relevant notification processing can be triggered.

8.10 CSV Export

The application provides a backend endpoint that generates booking data as CSV using Java I/O streams.



This demonstrates file and character-stream concepts while providing an administrative data-export facility.

8.11 Reflection

A restricted administrative metadata utility uses Java Reflection to inspect predefined entity classes and return information such as field names and field types.

The implementation does not allow arbitrary class loading from user input.

Screenshots / Results

Home Screen Interface:

SameNeed How It Works Services Open Groups All Portals Sign In Get Started

Local Service Pooling - Pune Communities

Group with neighbours. Save up to 40% on every service.

Why pay full individual rates when people in your apartment or neighborhood need the exact same maintenance? SameNeed clusters nearby demand so you get bulk quotes from certified technicians and negotiate together.

Select Locality Select Service Find Groups

Create a Service Request Browse Open Groups

Join as Provider

30-40% 100% Anonymous

GROUP FORMING Koregaon Park

Washing Machine Repair

Drum noise and drainage check across 3 flats in the same lane.

Unlock Tier: ₹450 / flat (~35% off regular ₹700)

3 / 5 Members Joined Join Group

OFFER RECEIVED Aundh

AC Summer Servicing

Complete filter deep clean and gas pressure top-up.

Fixit Pro Services
4.3 ★ · 8 yrs exp View Offer



How SameNeed Works

Four straightforward steps that turn single high-cost repairs into affordable group bookings.

1

Post Your Need

Specify the service category, describe your problem, and define your locality and preferred schedule.

2

Form a Group

Neighbours in your locality join your request. Your identity is kept private with anonymous aliases.

3

Receive & Negotiate

Verified local technicians bid on your group. Use group chat and counter-offers to reach the best price.

4

Confirm & Save

Accept the offer together, get the work completed at your agreed time, and pay the discounted group rate.

Built for Community Trust

Designed from the ground up to protect user privacy and guarantee service quality.



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Privacy Protected Chat

Every member in a group receives an anonymous alias (e.g. Member 02, Group Creator). Personal phone numbers and details remain hidden until bookings are confirmed.



Vetted Service Professionals

Contractors and technicians must verify business documentation, maintain minimum rating thresholds, and adhere to community service standards.



Transparent Bidding & Voting

All members see quotes, offer expirations, and negotiation rounds in real time. Offers can only be accepted when the group consensus is reached.

Popular Service Categories

Browse home and maintenance services frequently pooled by residential communities.

Home Appliances

Washing machines, refrigerators, microwaves & dishwashers.

[View Groups →](#)

Plumbing

Pipe leak detection, tap replacement & drain unblocking.

[View Groups →](#)

Electrical

Switchboards, fan installations & wiring safety checks.

[View Groups →](#)

AC & HVAC

Routine service, cooling troubleshooting & installation.

[View Groups →](#)

Cleaning

Deep house cleaning, sofa wash & kitchen sanitization.

[View Groups →](#)

Pest Control

Cockroach eradication, termite inspection & treatment.

[View Groups →](#)

Carpentry

Furniture repair, doors, latches & woodwork alterations.

[View Groups →](#)

Painting

Interior touch-ups, wall repairs & full home repainting.

[View Groups →](#)



Login Interface:

SameNeed

Create Account

I need a service

I provide services

Full Name

Phone

Your name

10-digit number

Email

you@example.com

Password (min 6 characters)

Your Locality / Area

e.g. Koregaon Park

Create Account

Already have an account? [Sign in](#)

Register Interface:

SameNeed

Create Account

I need a service

I provide services

Full Name

Phone

Your name

10-digit number

Email

you@example.com

Password (min 6 characters)

Business Name

Your business name

Service Area

e.g. Koregaon Park, Aundh

About Your Business

Brief description of your services...



User Dashboard

SameNeed

DashboardBrowse RequestsNew RequestBookings

Raviraaj Jain

Logout

Welcome back, Raviraaj Jain

Here's what's happening with your service groups.

0
My Requests

0
Groups Joined

0
Bookings

My Recent Requests

No requests yet
[Create your first request](#)

Browse All Requests

Recent Notifications

No notifications

SameNeed

DashboardBrowse RequestsNew RequestBookings

Raviraaj Jain

Logout

Browse Service Requests

Find a group to join or search for compatible requests to group with your own.

+ New Request

Category
Select Category

Service
All Services

Locality
e.g. Koregaon Park

Date
mm/dd/yyyy

Search

Washing Machine Repair

GROUP FORMING

Home Appliances • Koregaon Park

Washing machine is not draining after the wash cycle completes.

2026-09-21 at 10:00

3/10 members

₹500

Washing Machine Repair

GROUP FORMING

Home Appliances • Koregaon Park

Water is leaking from the bottom of the washing machine.

2026-09-21 at 10:00

3/10 members

₹600

Washing Machine Repair

Home Appliances • Koregaon Park

GROUP FORMING

Request Details

Problem: Washing machine is not draining after the wash cycle completes.

Date: 2026-09-21
Time: 10:00
Budget: ₹500
Posted by: Group Creator

Request Status

Group Info

Members: 3 / 10
Created: 18 Sept 2026, 12:12 am

Group Members

3 / 10

Group Creator

Member 02

Member 03

Join This Group

Offers & Negotiation



Service Provider Dashboard:

SameNeed

DashboardBrowse RequestsNew RequestBookings

Service Category

Select Category

Service

Select Service

Describe Your Problem

Explain what's wrong or what you need done...

Locality / Area

e.g. Koregaon Park

Preferred Date

mm/dd/yyyy

Preferred Time

--:-- --

Expected Budget (₹)

Optional

Max Group Size

10 members

Compatible Groups

Select a service, locality, and date to see if there are existing groups you can join instead.

SameNeed

DashboardGroup RequestsBookings

TestingLogout

Welcome, Testing

Manage your service offers and bookings.

0

Total Offers

0

Active Negotiations

0

Total Bookings

0

Active Jobs

Open Group Requests (Eligible for You)

No eligible requests at the moment

View All

My Recent Offers

No offers sent yet

Recent Bookings

No bookings yet

View All Bookings



Testing Approach

Testing was performed by testing individual modules as well as the complete end-to-end workflow.

10.1 Authentication Testing

- Valid registration
- Duplicate registration
- Valid login
- Invalid login
- Logout
- Unauthorized access

10.2 Service Request Testing

- Valid request creation
- Missing required fields
- Invalid values
- Request browsing
- Request joining
- Duplicate joining
- Full-group joining

10.3 Group Testing

- Group creation
- Group membership
- Anonymous aliases
- Maximum group size
- Leaving a group
- Unauthorized group access



10.4 Chat Testing

- Message sending
- Real-time message delivery
- Multiple connected users
- Message persistence
- Unauthorized WebSocket access

10.5 Negotiation Testing

- Provider offer creation
- Counter-offer
- Offer history
- Accept/reject
- Expired offer
- Member response

10.6 Booking Testing

- Booking creation
- Invalid booking conditions
- Status updates
- Completed booking
- Review eligibility

10.7 Admin Testing

- User management
- Provider verification
- Service management
- Report management
- Booking monitoring
- CSV export
- Reflection functionality

10.8 Database Testing

Database records were checked to verify that important operations were correctly persisted.



Tested records include:

- Users
- Requests
- Members
- Offers
- Negotiations
- Messages
- Bookings
- Reviews
- Notifications

Challenges Faced

11.1 Designing the Grouping Logic

One of the main challenges was deciding what should determine whether two customers belong to the same group.

The solution was to match users based on the structured service requirement rather than requiring identical problem descriptions.

11.2 Maintaining Anonymous Identities

The system needed to hide real customer identities while still maintaining the correct relationship between a user and their group membership.

This was handled by storing the actual user reference internally and assigning a request-specific display alias.

11.3 Real-Time Communication

Implementing real-time communication was more complex than normal HTTP request-response handling.

WebSocket was introduced to maintain active connections and deliver group messages without requiring page refreshes.

11.4 Negotiation State Management



Offers can move through different states such as pending, countered, accepted, rejected, expired, or cancelled.

Maintaining the negotiation history without losing previous offers required separate offer and negotiation records.

11.5 Background Processing

Offer expiry needed to happen automatically instead of relying on a user action.

A background worker was therefore implemented to periodically process expired offers.

11.6 Combining JDBC and JPA

Using both JDBC and JPA required careful separation of database responsibilities and configuration.

JDBC was kept as the primary persistence approach for most application operations, while JPA/Hibernate was used for selected entities to demonstrate ORM concepts.

11.7 Maintaining Role-Based Access

The application has three different user roles with different permissions. Ensuring that each role could access only the appropriate functionality required authentication and role-based filtering.

Learnings & Key Takeaways

The development of SameNeed provided practical experience with Java beyond basic console-based programming.

Java and OOP

The project provided practical use of:

- Classes and objects
- Constructors
- Encapsulation
- Inheritance



- Abstract classes
- Interfaces
- Method overloading and overriding
- Polymorphism
- Enumerations

The user-role hierarchy also provided a practical application of inheritance and polymorphism.

Exception Handling

Input and database operations required handling invalid conditions and runtime failures using Java exception-handling mechanisms.

Collections

Java collections were used for handling groups of objects, query results, messages, notifications, and other application data.

Multithreading

Background processing introduced practical use of Java threads for offer expiry and notification processing.

JDBC

The project provided experience with:

- Database connections
- Prepared statements
- Result sets
- CRUD operations
- Transactions
- SQL queries

JPA/Hibernate

The project introduced:

- Entity mapping
- Persistence units
- ORM
- Entity relationships



- JPQL/JPA-based persistence

Java I/O

CSV export provided practical experience with Java streams and file/data output.

Reflection

The administrative metadata utility provided practical experience with Java Reflection.

Web Development

The project provided experience with:

- HTTP
- Servlets
- Sessions
- Filters
- JSON APIs
- WebSocket communication
- WAR deployment
- Apache Tomcat

Software Design

A major learning was that application design should be based on actual requirements rather than adding technologies or classes only for the sake of complexity.

Future Enhancements

Possible future improvements include:

1. Online payment integration.
2. Email and SMS notifications.
3. Mobile application for customers and providers.
4. Improved provider availability management.
5. Location-based map integration.
6. Automated provider scheduling.
7. More advanced service-area matching.
8. Customer and provider cancellation policies.
9. Digital invoices and payment records.



10. Improved dispute-resolution workflow.
11. Provider performance analytics.
12. More detailed administrative reporting.
13. Push notifications.
14. Multilingual user interface.
15. Scalable deployment for larger numbers of concurrent users.

These enhancements are outside the scope of the current academic implementation but could be considered in a production-oriented version.

References

The following resources were used for understanding the technologies and concepts implemented in SameNeed:

1. **Oracle Java Documentation** — Java language, classes, inheritance, collections, exception handling, multithreading, I/O, Reflection, and JDBC.
2. **Jakarta EE Documentation** — Jakarta Servlet, WebSocket, and related Jakarta EE specifications.
3. **Hibernate ORM Documentation** — JPA/Hibernate entity mapping, persistence, and ORM concepts.
4. **MySQL Documentation** — MySQL database administration, SQL syntax, tables, constraints, and JDBC connectivity.
5. **Apache Maven Documentation** — Maven project configuration, dependency management, and build lifecycle.
6. **Apache Tomcat Documentation** — Web application deployment and Jakarta Servlet container configuration.
7. **jBCrypt Documentation** — BCrypt password hashing implementation for Java.
8. **Jackson Documentation** — JSON serialization and deserialization using Jackson Databind.
9. **VIT Bhopal University — VITYarthi Course Requirements** — Project requirements, documentation structure, implementation expectations, and evaluation criteria.
10. **Course Materials and Classroom Notes** — Java programming, OOP, JDBC, JPA, multithreading, collections, I/O, and web application concepts covered during the course.



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